Technical Report:

1. Scenario definition

Over the past 20 years, the MARSIS antenna has experienced a series of anomalies that have affected the quality of its observations, significantly reducing the number of observations usable for scientific purposes. Considering the occurrence of these anomalies is crucial in observation planning to mitigate their impact and maximize the number of high-quality observations.

We chose to represent the model using a Decision-Dependent Evaluation Process (DEP), treating potential anomaly occurrences over the time horizon as scenarios. There is no strict rule for defining these scenarios, although scenario definition is widely acknowledged as a key component in stochastic programming.

Rather than working with a limited set of discrete outcomes with known probabilities, we deal with a range of phenomena that affect observation quality according to probabilistic distributions. Therefore, we developed a scenario generator capable of theoretically producing an infinite number of possible outcomes. The goal is to generate a sufficiently large set of equiprobable scenarios that closely approximate future conditions.

Each scenario is created as a combination of various phenomena in all possible configurations. We identified three distinct types of phenomena: machine learning (ML) prediction errors, machine failures, and solar-induced blackouts. For comprehensive modeling, we assume these phenomena are linearly independent, given their different causes.

2.1 ML Error

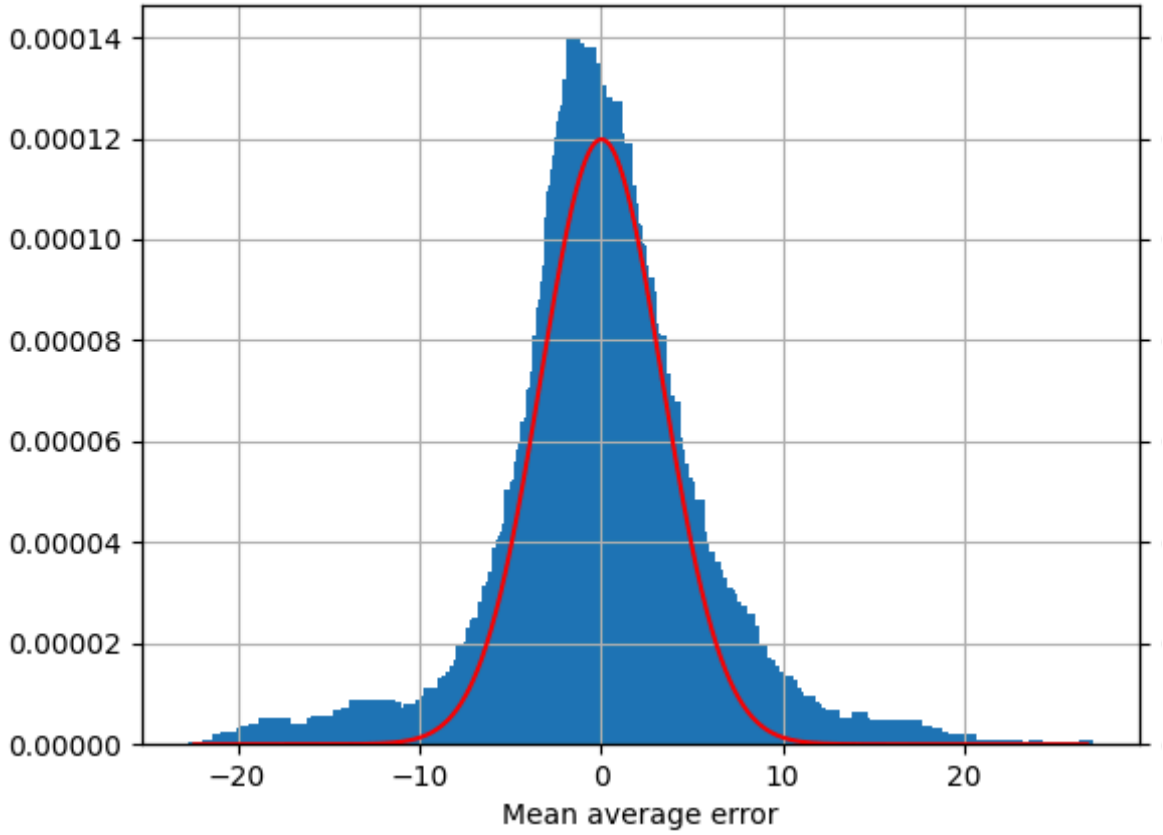
We trained a machine learning algorithm to predict the quality of future observations. In our previous publication, we treated these predictions as exact during model evaluation. However, the model has a mean absolute error (MAE) of 3.37. This means that, on average, the absolute difference between predicted and actual values is 3.37, though individual errors can vary significantly. In our case, the observed range is from -22.36 to 26.72.

We analyzed the prediction errors from 2,076,917 echoes using 10-fold cross-validation, recording the frequency of each error rounded to three decimal places. Due to the large dataset, we grouped the errors at the third decimal and calculated the relative frequency of each group by dividing by the total number of echoes.

When plotting the sorted error groups, the distribution closely resembles a scaled Gaussian distribution (that is, a normal distribution multiplied by a scaling factor). In our case, the scale factor is 0.001, corresponding to the granularity of the error bins. This approximation

results in an average error of just 0.0004% in probability, indicating that this is a consistent trend in the multi-layer perceptron's predictions.

For each future observation, we simulate a prediction error by sampling a random number between 0 and 1. Using the cumulative distribution of the scaled Gaussian, we apply the sampled error to the predicted Peak Power value.



(Figure: Distribution of prediction errors percentage for ML model.)

2.2 Machine Failures

MARSIS acquires a series of echoes in rapid succession, which are later combined into a single observation. Occasionally, it misses one or more echoes or processes them incorrectly. These failures usually appear as black stripes in the data, rendering the observations unusable.

While such failures were common in the early operational years of Mars Express, they have become rare in recent times. Nevertheless, they must be accounted for in realistic scenario generation. Based on recent data, we fixed the failure rate at 0.005%, derived from the only machine failure observed among the last 20,000 observations.

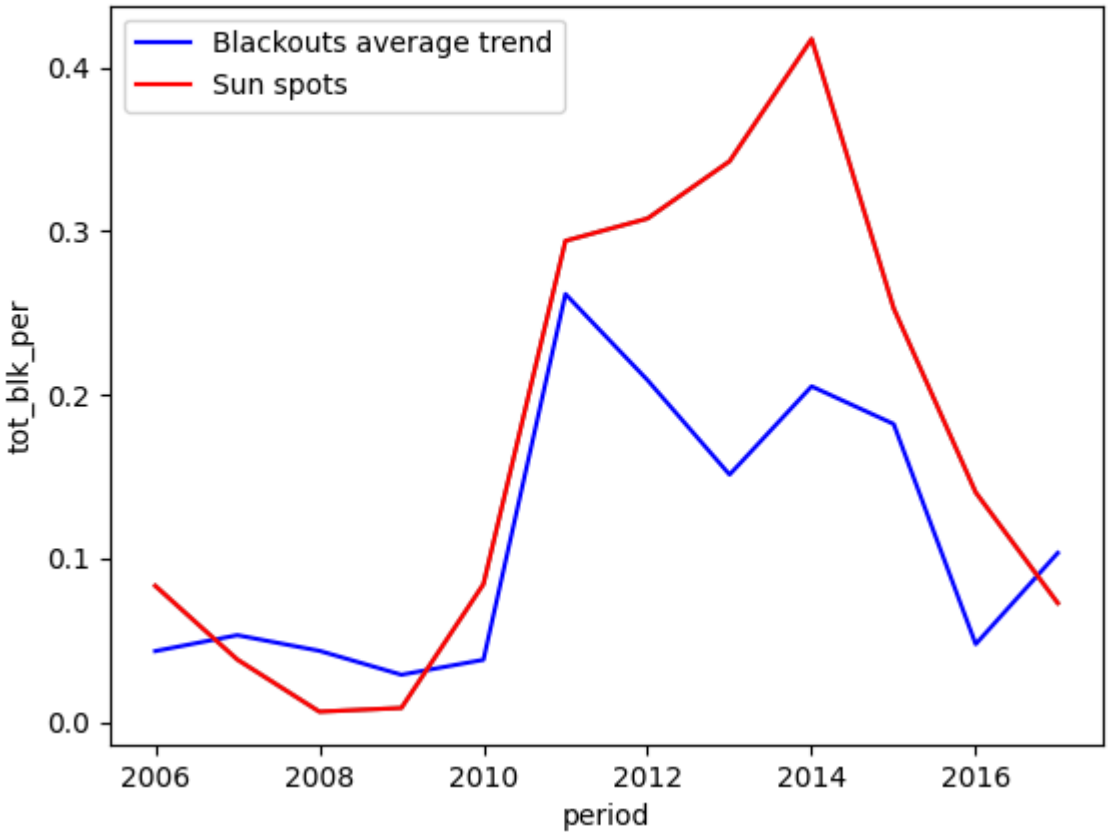
2.3 Sun Blackouts

A significant number of degraded observations can be traced back to solar activity. Events such as interplanetary coronal mass ejections or solar flares can saturate Mars' ionosphere with charged particles, which absorb the MARSIS signal.

Although solar behavior cannot be predicted with precision, its influence correlates with the solar cycle, an approximately 11-year pattern often monitored through sunspot activity. Sunspots—dark regions on the solar surface caused by intense magnetic activity—serve as proxies for solar phenomena. The International Sunspot Number is the most widely used metric, with reliable records dating back to 1818.

By aggregating blackout data from 2006 to 2017, we observed a pattern that aligns with sunspot numbers (expressed as a percentage of the historical maximum). This trend features two peaks, associated with the so-called Gnevyshev gap, a brief (1–2 year) increase in sunspot numbers that typically occurs within each cycle.

Given the current solar cycle and available data, we simulate blackout behavior by replicating historical patterns. For each scenario, we generate spikes based on the observed distribution of past occurrences. Around each spike, we model total blackouts with a double Gaussian distribution and partial blackouts with a single Gaussian centered between the two peaks.



(Figure: Correlation between blackout occurrences and sunspot number.)

2. Stochastic model definition

MOSP aims at the generation of observation schedules for the radar MARSIS in order to maximize the coverage of areas of interest on Mars and collect high-quality data. We focus on the coverage of the South Pole, since in this area lakes of underground water have been retrieved. Specifically, in MOSP we have a set of orbits of MEX around Mars in a specific planning horizon D. Due to memory limitations, we can select only a single 134 second observation per day. The surface of Mars has been discretized in a set P of Point of Interests (PIs). Each PI has a weight defined by the stakeholders, which reflects its priority in the planning process. We are given the list O of future observation opportunities, where each observation covers a set of PIs with a predicted quality that belongs to a defined set Q. Ideally, the quality depends only on the location and time of the observation. In practice, it also depends on stochastic events.

To deal with this, we defined a set of scenarios S representing the realization of different events, based on the probability of their occurrence. These scenarios are then used for the development of a scenario-based planning approach. Each scenario takes into consideration the probability of a solar event that may cause a total blackout corresponding to a null quality, or a partial blackout, corresponding to a reduced quality compared to what we predicted. The probability is derived from the study of past solar events and the impact on MARSIS

observations. Moreover, each scenario considers the probability of an instrument malfunction and the error in the predictions generated by the machine learning approach.

Table 1: Parameters

Input	Description
$d(o)$	day in which the observation $o \in O$ is performed in D
a_{op}	1 if $o \in O$ contains the PI $p \in P$, 0 otherwise
$\bar{q}_p$	max quality of past observations of $p \in P$
w_p	weight of each PI $p \in P$ defined by the stakeholders
$v_q \in V$	value of the quality $q \in Q$
π_s	probability of occurrence of the scenario $s \in S$
V	set of all possible quality values
$O^s(p,q)$	set of observations of p with quality q in the scenario s
N	max number of daily observations (in our study $N = 1$)

Most future observation data can be defined in advance (e.g., latitude, longitude and altitude of the spacecraft) but the quality cannot, even considering the absence of Sun interference. To overcome this problem, an ML algorithm was developed for the prediction of the quality (Ferrari et al. 2024). We adopt the same algorithm in our work. It consists of a neural network, which outperformed other ML algorithms in terms of accuracy thanks to its ability to detect the non-linearity of the complex Mars environment. Then, we define a DEP to find an optimal solution considering all the generated scenarios.

The mathematical model starts from the deterministic formulation already presented in (Ferrari et al. 2024) and based on (Church and ReVelle 1974). The variables involved are two. The first is y_{spq}, which is used to identify if a specific PI in $p \in P$ is observed with quality $q \in Q$ in a scenario $s \in S$. The second is x_o, which indicates if an observation $o \in O$ is scheduled. Taking into account also the parameters as defined in Table 1, the resulting DEP model is:

$$\max \quad z = \sum_{s \in S} \sum_{p \in P} \sum_{q \in Q: q \geq \overline{q}_p} \pi_s \cdot (v_q - v_{\overline{q}_p}) \cdot w_p \cdot y^s_{pq} \tag{1}$$

$$s.t. \quad y^s_{pq} \leq \sum_{o \in O^s(p,q)} a_{op} \cdot x_o \qquad p \in P, q \in Q, s \in S \tag{2}$$

$$\sum_{q \in Q} y^s_{pq} \leq 1 \qquad p \in P, s \in S \tag{3}$$

$$\sum_{o \in O(d)} x_o \leq N \qquad d \in D \tag{4}$$

$$y^s_{pq} \in \{0, 1\} \qquad p \in P, q \in Q, s \in S \tag{5}$$

$$x_o \in \{0, 1\} \qquad o \in O \tag{6}$$

The objective function (1) considers the maximum gain of the observations, given any possible scenario s ∈ S with probability πs, weighting the binary variable yspq with the importance wp of the PI and the quality gain which is calculated as the difference between the values of the predicted quality class v_{sq} and the initial value $\underline{v}$qp.

The constraints can be interpreted as follows:

- Constraints (2) ensure that a PI p ∈ P is considered covered with quality q ∈ Q only if it is included in at least one selected observation o ∈ O_s(p, q);
- Constraints (3) prevent the model from collecting a profit q ∈ Q from a PI more than once
- Constraints (4) limit the number of selected observations to the maximum allowed N for each day d ∈ D;
- Constraints (5) and (6) define the domain of the variables involved.

3. Testing

The algorithms are coded in Python and run on a QEMU Virtual CPU version 2.5 + at 2.10 GHz and 16 GB RAM, while the models were solved using Gurobi 12.

The testing was made using 653 orbits, 113,651 observation opportunities, in the period of 213 days (2023-2025) considering a validated dataset of observation performed at 4 MHz on Mars and having experienced any known blackouts as.

The following settings where implemented:

- Discretization of the grid = 3, 5, 10 km
- Scenarios= 5, 10, 30, 50
- Timelimit = 1 hour

10 km - 15,681 PI

G	S	UB	Obj	Time	GAP	Obj	Time	GAP	Obj	Time	GAP
			BENDERS			DEP			MATH		
10	5	2,358.05	2,250.85	TL	4.55%	**2,358.05**	**56**	**0.00%**	2,357.90	2,534	0.01%
10	10	2,301.53	2,201.23	TL	4.36%	**2,301.53**	**180**	**0.00%**	2,300.87	TL	0.03%
10	30	2,260.90	1,904.72	TL	15.75%	**2,260.90**	**1,473**	**0.00%**	2,259.98	TL	0.04%
10	50	2,254.17	2,011.84	TL	10.75%	**2,254.17**	**3,408**	**0.00%**	2,250.28	TL	0.17%

5 km - 62,736 PI

G	S	UB	Obj	Time	GAP	Obj	Time	GAP	Obj	Time	GAP
			BENDERS			DEP			MATH		
5	5	8,917.00	8,624.35	TL	3.28%	**8,917.00**	**2,553**	**0.00%**	8,914.73	TL	0.03%
5	10	9,480.29	8,477.00	TL	10.58%	2,347.60	TL	75.24%	**8,755.79**	**TL**	**7.64%**
5	30	9,319.53	8,057.48	TL	13.54%	OOM			**8,487.30**	**TL**	**8.93%**
5	50	9,295.19	7,806.67	TL	16.01%	OOM			**8,078.27**	**TL**	**13.09%**

3 km - 174,283 PI

G	S	UB	Obj	Time	GAP	Obj	Time	GAP	Obj	Time	GAP
			BENDERS			DEP			MATH		
3	5	26,771.29	24,011.55	TL	10.31%	5,625.95	TL	78.99%	**24,542.51**	**TL**	**8.33%**
3	10	26,083.03	23,098.08	TL	11.44%	OOM			**23,713.69**	**TL**	**9.08%**
3	30	25,605.88	**21,941.33**	**TL**	**14.31%**	OOM			18,945.56	TL	26.01%
3	50	25,527.07	**22,318.64**	**TL**	**12.57%**	OOM			10,417.89	TL	59.19%

The results show a failing of the Deterministic Equivalent Problem at the growth of complexity, when the Matheuristic first and the Benders later starts to outperform the Model at TL, not ending the memory given.

Detailed computational results will be available after the end of the test and the submission of the paper.